# Capstone Expo

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

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WORKBOOK · 1 OF 21

## Recall and choose your starting point

> Record your starting explanation for Capstone Expo before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Improve and Challenge

Without opening the old lesson, explain one key idea from Improve and Challenge. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Build the Recommender

Without opening the old lesson, explain one key idea from Build the Recommender. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Agents and Tools

Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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WORKBOOK · 2 OF 21

## Practise with fading help

> Complete the supported Capstone Expo trace and record the first step you can explain without help.

Use W2: Check report consistency for Defending the data application. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 21

## Individual reasoning check

> Check the proposed Capstone Expo explanation against the supplied facts before discussing it.

Use the worked case for Defending the data application. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 21

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

**Independent case: Defend a new case independently**

Limit is 10. A has minutes 10, fun 2, learning 4; B has 9,3,2; C has 11,5,5; D has 8,"4",2. Validate, rank, give counts, explain the winner, then reverse the **input** order. Present without hints.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh **transfer**.

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WORKBOOK · 5 OF 21

## Your learning target

> Connect the Capstone Expo target to a trace and a justified conclusion.

Present a recommender that gets 8/10 final requests right. One failure invents a title; another violates a hard page limit. Is '**80%**, ready for everyone' justified? Propose release criteria.

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WORKBOOK · 6 OF 21

## Solve the given case

> Solve the supplied Capstone Expo case and show the deciding evidence.

An expo claim says the recommender knows everybody's favourite book after twelve tests.

Rewrite the defence.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

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WORKBOOK · 7 OF 21

## Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Your demo recommends the right book once. An audience member asks why they should trust it. Name **evidence** and one **limitation** you should present.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

**Week 36 Test and reflect**

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WORKBOOK · 8 OF 21

## Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

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## Project checkpoint

> Justify what Capstone Expo adds to Transparent Book Recommender and keep the evidence.

Final **evidence bundle**: brief, data, scoring logic, tests, improvement record, and product card.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

## Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

## Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Present a recommender that gets 8/10 final requests right. One failure invents a title; another violates a hard page limit. Is '**80%**, ready for everyone' justified? Propose release criteria.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

## W1 Assemble evidence

> Assemble evidence. Apply the supplied Capstone Expo facts and keep your reasoning.

List the files or records a peer needs to reproduce the chooser result.

**My reasoning and evidence**

WORKBOOK · 13 OF 21

## W2 Check report consistency

> Check report consistency. Apply the supplied Capstone Expo facts and keep your reasoning.

A report says supplied 5, accepted 4 and rejected 2. Identify the inconsistency.

**My reasoning and evidence**

WORKBOOK · 14 OF 21

## W3 Repair a claim

> Repair a claim. Apply the supplied Capstone Expo facts and keep your reasoning.

A presentation says the project proves students learn better. What claim does a successful rule-based demo support instead?

**My reasoning and evidence**

WORKBOOK · 15 OF 21

## W4 Defend a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Defend a new case independently. Apply the supplied Capstone Expo facts and keep your reasoning.

Limit is 10. A has minutes 10, fun 2, learning 4; B has 9,3,2; C has 11,5,5; D has 8,"4",2. Validate, rank, give counts, explain the winner, then reverse the **input** order. Present without hints.

**My reasoning and evidence**

WORKBOOK · 16 OF 21

## W5 State the next evidence need

State the next evidence need. Apply the supplied Capstone Expo facts and keep your reasoning.

After passing the new case, what remains unproven and what is a reasonable next step?

**My reasoning and evidence**

WORKBOOK · 17 OF 21

## Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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**WORKBOOK · 18 OF 21**

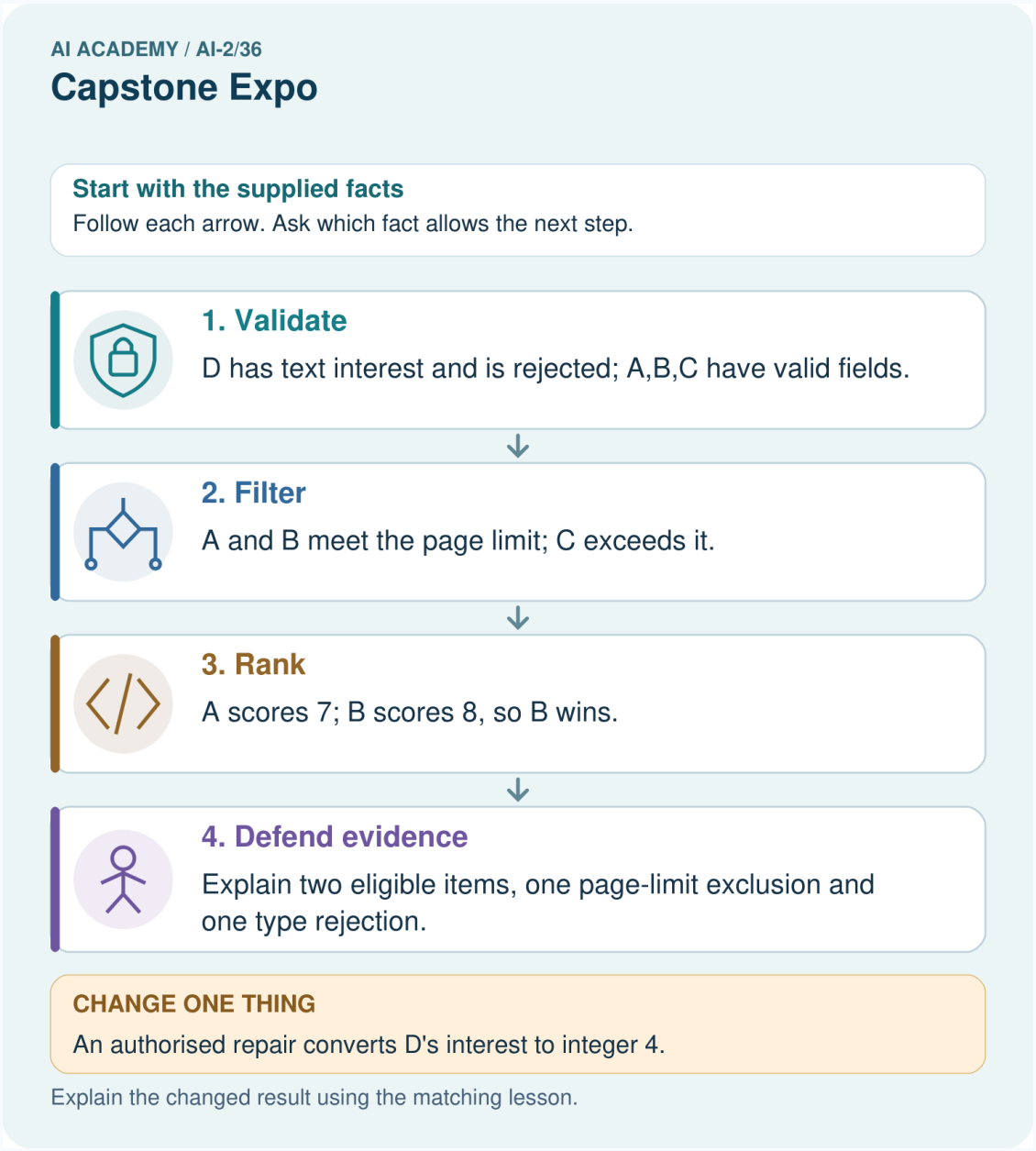
## Pictorial case practice: Capstone Expo

> A fictional team presents its transparent book recommender on a reserved catalogue. Explain the labelled step before checking the changed condition.

A fictional team presents its transparent book recommender on a reserved catalogue.

Request maximum 120 pages; score=2×interest+illustrated, where illustrated is 1 for True and 0 for False; tie by ID.

A pages120 interest3 True; B pages100 interest4 False; C pages121 interest5 True; D pages90 interest"4" True. Numeric fields require exact integers.




Follow the labelled picture with the matching explanation. Open the vector diagram

Give the original winning ID, its score, and counts of eligible, excluded and invalid rows.

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Present the Transparent Book Recommender’s code, baseline, full-case traces, reserved results and **limitation**s, without claiming fictional results as pupil work.

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## Project portfolio milestone

> Keep this week’s input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Present the Transparent Book Recommender’s code, baseline, full-case traces, reserved results and **limitation**s, without claiming fictional results as pupil work.

| Evidence to keep | My record |
| --- | --- |
| This week’s input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional team presents its transparent book recommender on a reserved catalogue.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Explain two eligible items, one page-limit exclusion and one type rejection.

1. Defend **evidence**

2. Validate

3. Filter

4. Rank

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Defend **evidence** Explain two eligible items, one page-limit exclusion and one type rejection.

A presentation should let someone rebuild the result from the stated contract. A's equality is accepted; D is not silently converted; C's high preference score cannot override the page requirement.

Reversing rows must preserve B because it has the unique highest eligible score. Keep the reserved case distinct from practice **demonstration**s.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: An authorised repair converts D's interest to integer 4. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. D becomes eligible and scores 9, so it wins.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: D becomes eligible and scores 9, so it wins. The **evidence**-backed **input** repair changes validation and then ranking; record this as a new case/version.

### Repair a missing building block

Review Improve and Challenge (ai-2/35). Goal: Repair a specific observed failure. Short practice: Why are tests at both 100 and 101 needed?. Return to this week and explain what you changed.

Review Build the Recommender (ai-2/33). Goal: Implement a deterministic recommender. Short practice: Explain why B does not win after D is excluded even though it has fewer pages..

Return to this week and explain what you changed.

Review Agents and Tools (ai-2/30). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Can the assistant accept Trace B because its final state is done?.

Return to this week and explain what you changed.

### Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

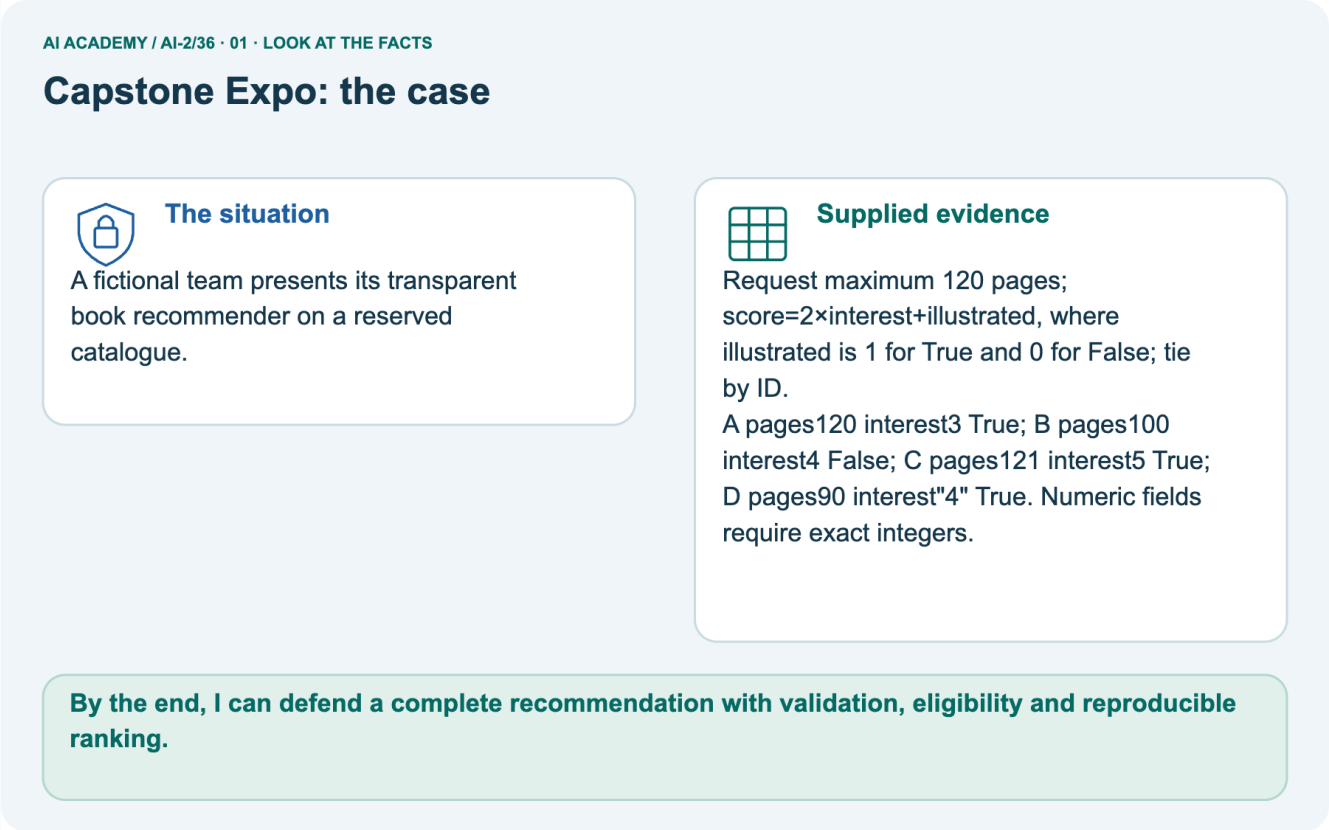
**What I repaired, the evidence I kept and the next test**

WORKBOOK · 21 OF 21

# Visual concept journey

> Explain each Capstone Expo picture from its labels, then record a relevant limit.

### 01. Capstone Expo: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

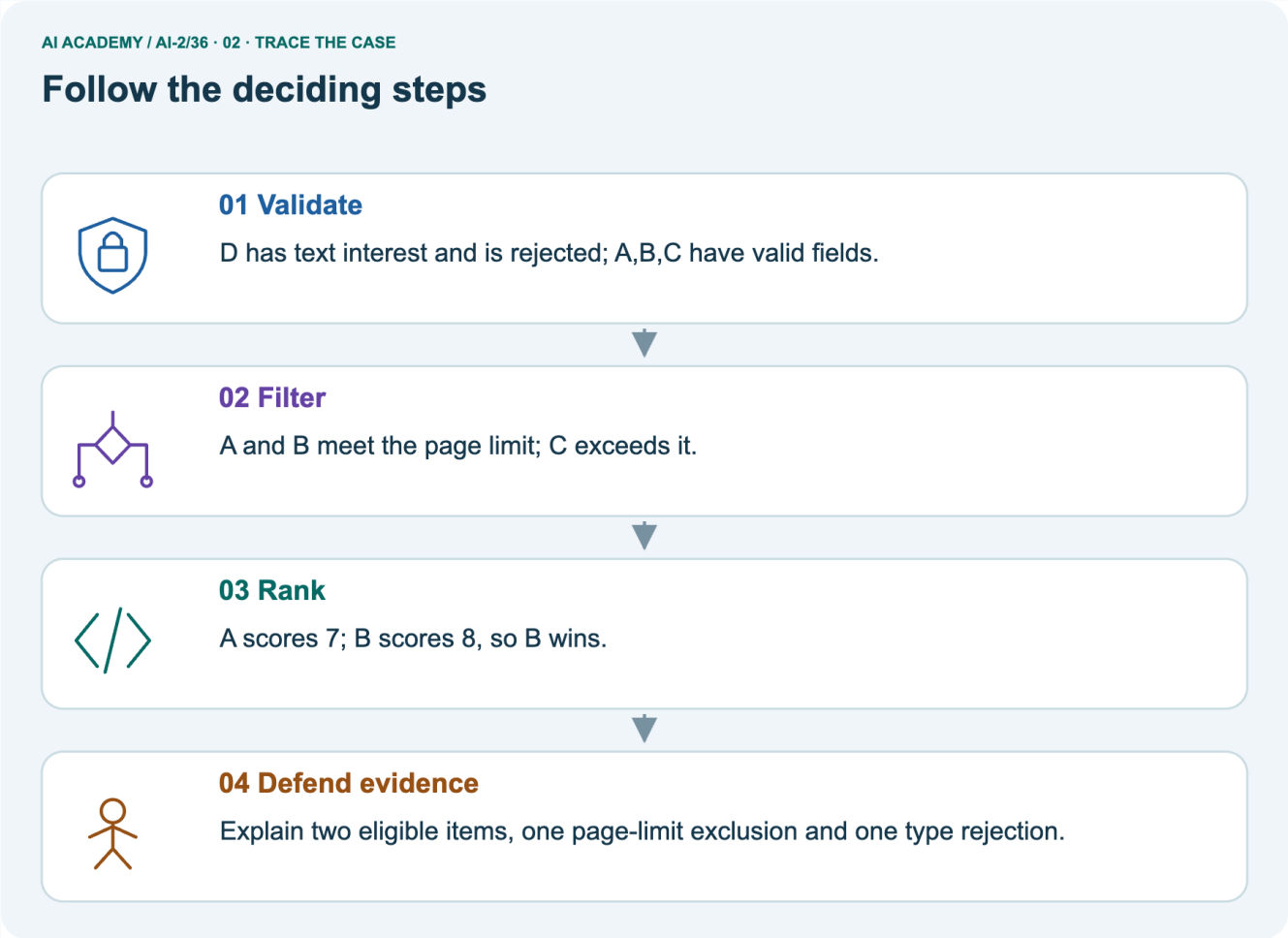
A fictional team presents its transparent book recommender on a reserved catalogue. Supplied facts: Request maximum 120 pages; score=2×interest+illustrated, where illustrated is 1 for True and 0 for False; tie by ID.

A pages120 interest3 True; B pages100 interest4 False; C pages121 interest5 True; D pages90 interest"4" True. Numeric fields require exact integers.

> Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps



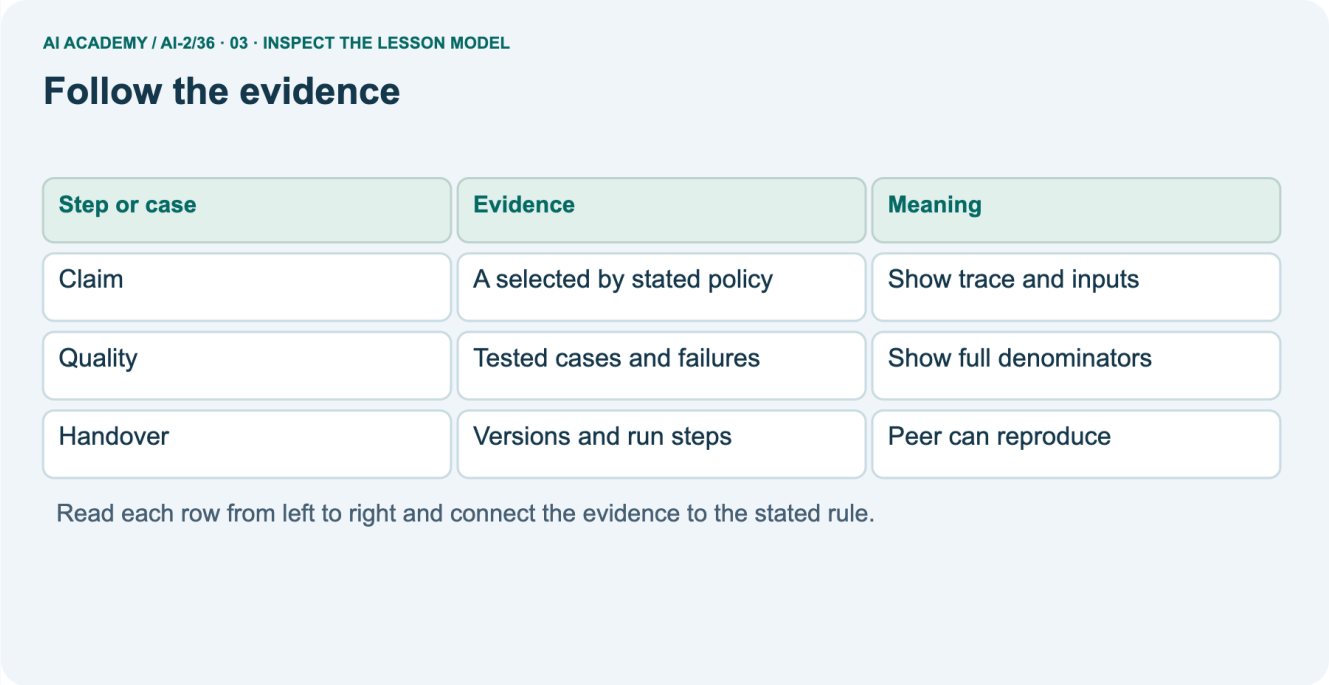

Follow the labelled picture with the matching explanation. Open the vector diagram

Validate: D has text interest and is rejected; A,B,C have valid fields. → Filter: A and B meet the page limit; C exceeds it. → Rank: A scores 7; B scores 8, so B wins. → Defend **evidence**: Explain two eligible items, one page-limit exclusion and one type rejection.

> Point to each numbered stage and explain which recorded **input** it uses.

**My picture explanation, deciding evidence and one limit**

## 03. Follow the evidence



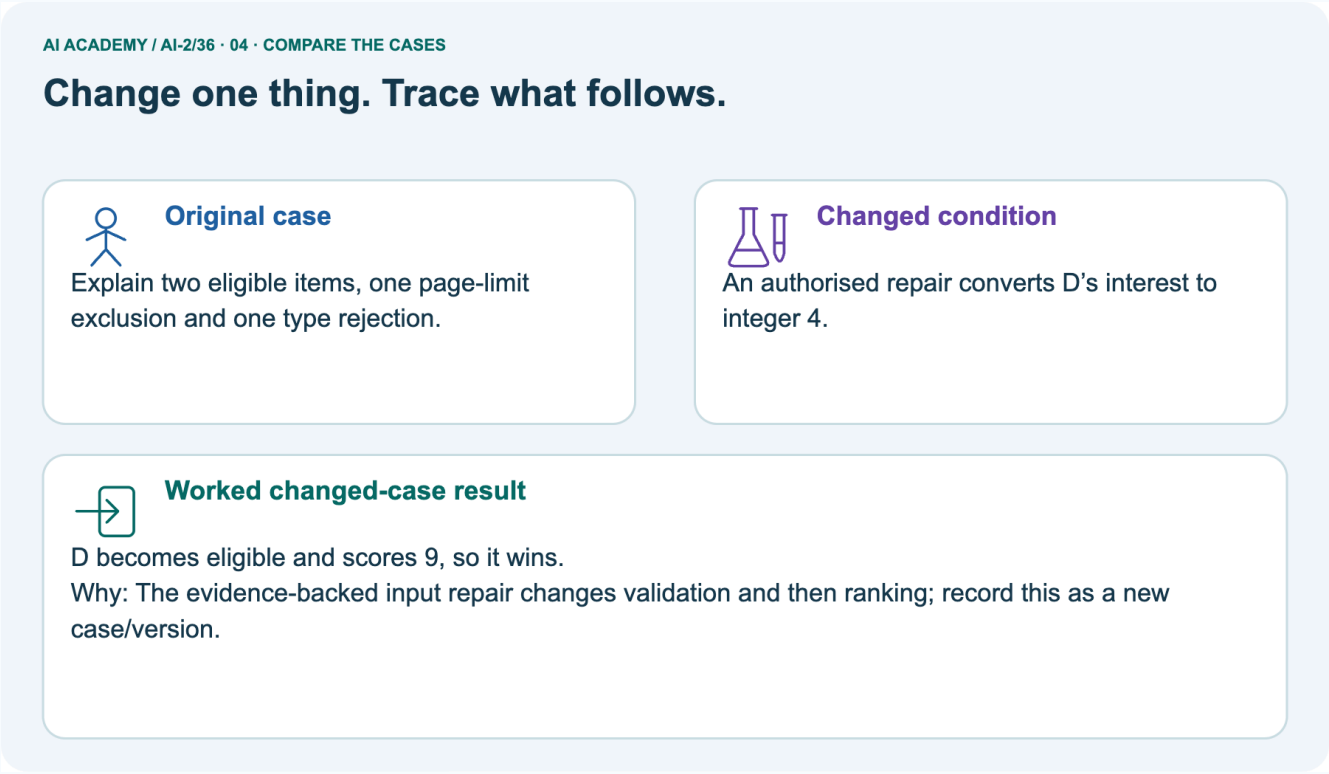

Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Claim | A selected by stated policy | Show trace and **input**s. Quality | Tested cases and failures | Show full denominators. **Handover** | Versions and run steps | Peer can reproduce

Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

### 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

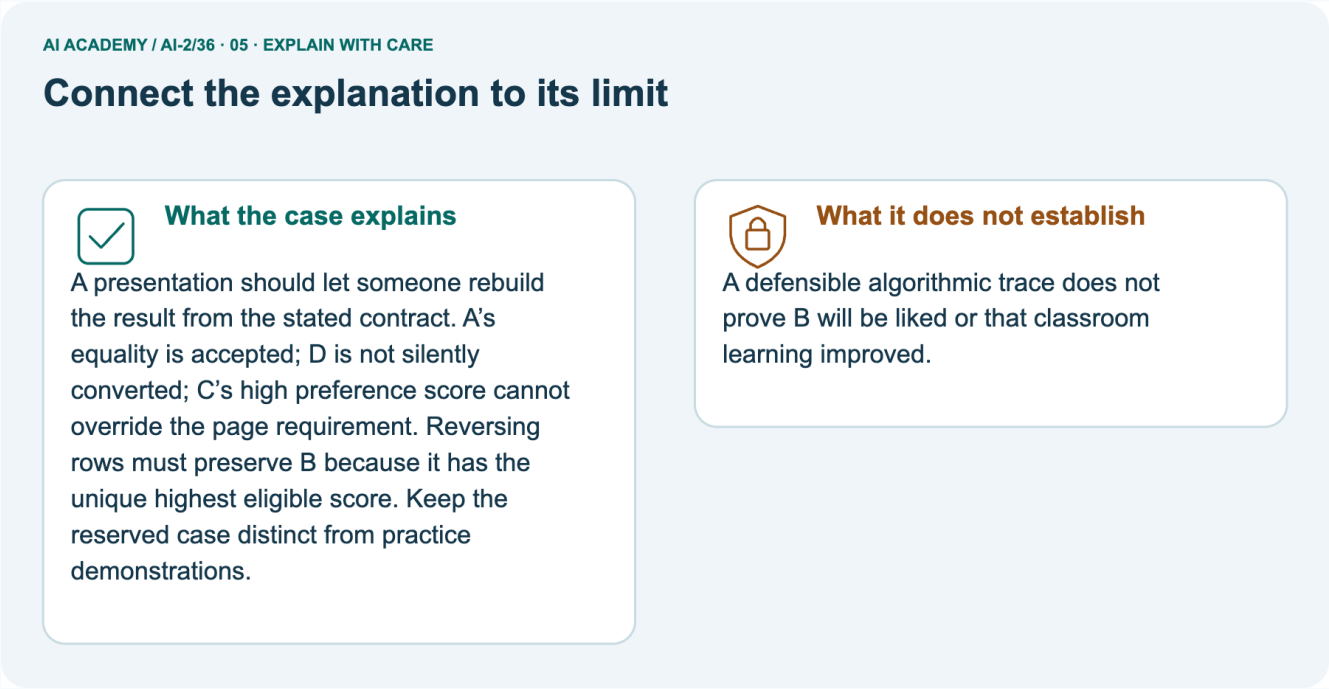
Original case: Explain two eligible items, one page-limit exclusion and one type rejection. Changed condition: An authorised repair converts D's interest to integer 4.

Expected result: D becomes eligible and scores 9, so it wins. Reason: The **evidence**-backed **input** repair changes validation and then ranking; record this as a new case/version.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

A presentation should let someone rebuild the result from the stated contract. A's equality is accepted; D is not silently converted; C's high preference score cannot override the page requirement.

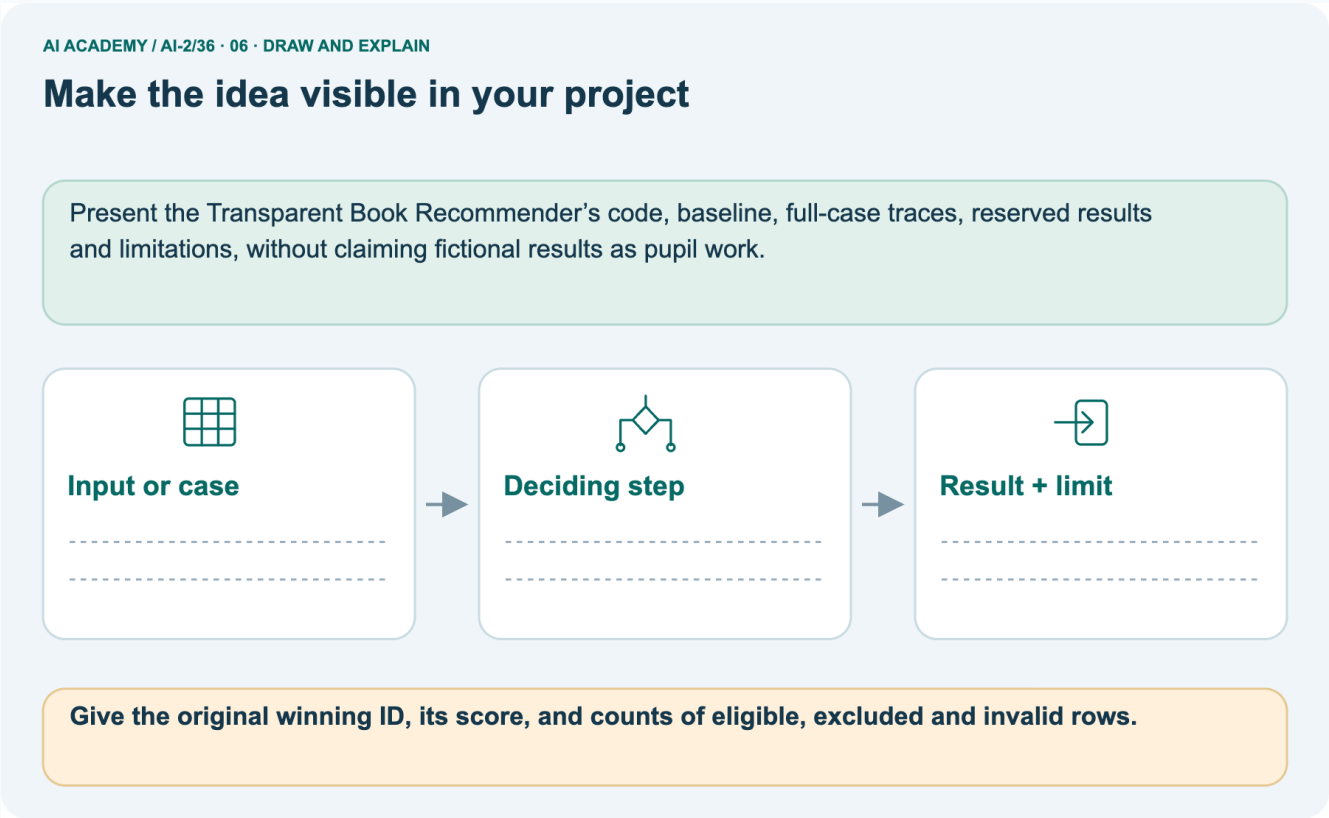
Reversing rows must preserve B because it has the unique highest eligible score. Keep the reserved case distinct from practice **demonstration**s.

Limit: A defensible algorithmic trace does not prove B will be liked or that classroom learning improved.

Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Present the Transparent Book Recommender's code, baseline, full-case traces, reserved results and **limitation**s, without claiming fictional results as pupil work. Draw your **input**, deciding step, result and limit.

Supported question: Give the original winning ID, its score, and counts of eligible, excluded and invalid rows.

Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.